

Correlation

Introduction:

You have previously learned in Statistics how to describe a set of data representing a phenomenon using some statistical scales such as the scales of the central tendency, dispersion and the coefficient of variation. In this lesson, you are going to learn how to describe the items of two different phenomena in accordance to the relation between both. In other words, if one of the two variables has changed in a certain direction (by increasing or decreasing), the other variable leans to change in a certain direction by increasing or decreasing, too. In this case, the correlation is called a direct correlation. If one of the two variable changes in the direction of increasing and the other changes in the direction of decreasing and vice versa, the correlation in this case is called the inverse correlation.

meditate the following examples and record your observations:

- 1- The relation between the side length and the area of the square.
- 2- The relation between the blood pressure and the age.
- 3- The increase in the price of an item and the demands of buying it.
- 4- The decrease of temperature and the demand to consume the fuel.
- 5- The relation between rising of the sea level and the rise of the temperature.

From the examples above, we observe that:

a The two correlated variables change in the same direction. I.e. the increasing or decreasing of a variable leads to the increasing or decreasing of the other as in examples 1,2 and 3. It is said the correlation between both variables is positive (direct).

In examples 4 and 5, we notice that two correlated variables inversely change. In other words, the increasing or decreasing of one variable leads to decreasing or increasing of the other variable. hence, it is said that the correlation between them is negative (inverse).

Definition of Correlation:

Correlation is a statistical method by which the degree and the type between two variables can be determined.

Scatter Diagram Definition:

Scatter Diagram is a graphical representation of a number of the ordered pairs (x,y) to describe the relation between two variables.

If we denote the first phenomenon by the symbol (x) and the second phenomenon by the symbol (y) , then the next diagrams illustrate the relation between x and y which show the scatter diagram.

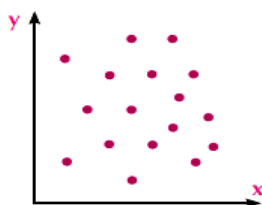


Figure 1
There is no correlation

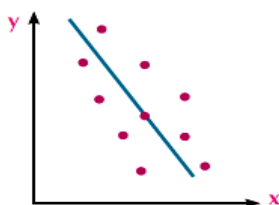


Figure 2
There is an inverse linear correlation

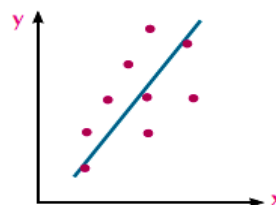


Figure 3
There is a direct linear correlation

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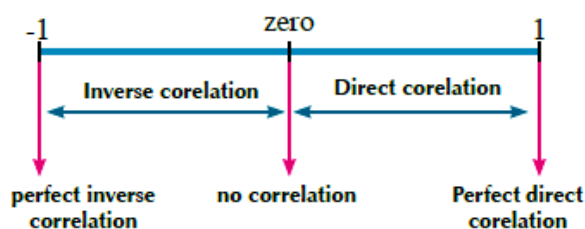
Latter we will discuss how to get this line?

Correlation Coefficient:

The correlation coefficient is denoted by the symbol (r). It is a relatively quantitative scale measuring how much the two variables are correlated where $-1 \leq r \leq 1$. It is said the correlation is perfect direct if the correlation coefficient $r = 1$, is perfect inverse if the correlation coefficient $r = -1$ and the correlation is nihilistic when $r = 0$.

We notice that:

The nearer the value of the correlation coefficient to 1 is, the stronger the direct correlation between the two



variable is and the nearer its value to zero is, the weaker the direct correlation is and we will say the same in the inverse correlation. The opposite figure shows that.

Pearson's Correlation coefficient:

Suppose we have a set of (n) people and we get data from those people about the values of two variables x and y . The data we get ought to be on the form:

The values of first variable x : $x_1, x_2, x_3, \dots, x_n$

The values of second variable y : $y_1, y_2, y_3, \dots, y_n$

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If we denote the correlation coefficient by the symbol (r), then Pearson's correlation coefficient between the two variables x and y or the linear correlation coefficient can be found from the relation:

$$r = \frac{n \sum xy - (\sum x \times \sum y)}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$

Where: " $\sum$ " is the symbol of summation notation and read as the sum.

EXAMPLE 1►

The next table shows the marks of 10 students in the history and geography subjects.

history x	75	80	93	65	87	71	98	69	84	78
geography y	82	78	86	72	91	80	95	73	89	74

Calculate the Pearson's correlation coefficient between the two variables x and y and identify the type of the correlation.

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Solution:

We form the next table:

x	y	x ²	y ²	x y
75	82	5625	6724	6150
80	78	6400	6084	6240
93	86	8649	7396	7998
65	72	4225	5184	4680
87	91	7569	8281	7917
71	80	5041	6400	5680
98	95	9604	9025	9310
69	73	4761	5329	5037
84	89	7056	7921	7476
78	74	6084	5476	5772
Σx =800	Σy =820	Σx^2 =65014	Σy^2 =67820	Σxy =66260

$$\begin{aligned}\therefore r &= \frac{n \Sigma xy - (\Sigma x \times \Sigma y)}{\sqrt{n \Sigma x^2 - (\Sigma x)^2} \sqrt{n \Sigma y^2 - (\Sigma y)^2}} \\ \therefore r &= \frac{10 \times 66260 - (800 \times 820)}{\sqrt{10 \times 65014 - (800)^2} \sqrt{10 \times 67820 - (820)^2}} \\ &= \frac{6006}{\sqrt{10140} \sqrt{5800}} \simeq 0.8606 \quad \text{The correlation is direct .}\end{aligned}$$

EXAMPLE 2➤

Find the Pearson's correlation coefficient between the two variables x and y and identify its type if.

$$\Sigma x = 68$$

$$\Sigma x^2 = 620$$

$$\Sigma y = 36$$

$$\Sigma y^2 = 204$$

$$\Sigma xy = 348$$

$$n = 8$$

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Solution:

$$\therefore r = \frac{n \sum xy - (\sum x \times \sum y)}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}}$$
$$\therefore r = \frac{8 \times 348 - (68 \times 36)}{\sqrt{8 \times 620 - (68)^2} \sqrt{8 \times 204 - (36)^2}} = \frac{336}{\sqrt{336} \sqrt{336}} = 1$$

The value of the correlation coefficient (+1) means that the relation between the two variables x and y is perfect direct.

TRY TO SOLVE ➤

Find the Pearson's correlation coefficient between the two variables x and y and identify its type if.

$$\begin{aligned}\sum x &= 92 \\ \sum x^2 &= 1100\end{aligned}$$

$$\begin{aligned}\sum y &= 36 \\ \sum y^2 &= 204\end{aligned}$$

$$\begin{aligned}\sum xy &= 372 \\ n &= 8\end{aligned}$$

Spearman's Rank Correlation Coefficient

Think and discuss A statistician has studied the relation among the degrees of two school subjects for seven students and has recorded the results in the next table:

Subjects 1	Weak	Pass	Weak	good	Weak	excellent	Very good
Subjects 2	Weak	Pass	good	Pass	Weak	Very good	Pass

can you help the statistician , to realize the relation between those two subjects and find the correlation coefficient between them?

We cannot use Pearson's correlation coefficient in "Think and discuss" since it depends on the quantitative (numerical) data only. In this case, the data are descriptive (as in the previous title). another correlation coefficient can be used Spearman's rank correlation coefficient. It gives a measure to the correlation in both the descriptive and quantitative data which are ordered as in the previous title. This coefficient relies on the order of the variable values taking into consideration the ascending and descending order. Then, we use the next relation:

$$r = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}$$

Where D is the difference between the order of the two variables x and y and n is the number of the values of each variable.

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Notice that:

- Spearman's correlation coefficient can be calculated whether the data are quantitative or descriptive, whereas Pearson's correlation coefficient can be calculated only if the variables are quantitative.
- Spearman's rank correlation coefficient is characterized with its easiness even the data are not ordered.
- The Spearman's correlation coefficient does not regard the difference of the numbers when the ranks are calculated and though, it is less accurate.

Example 3➤

Find Spearman's rank correlation coefficient in the previous "Think and discuss" and identify its type.

Solution:

In this example, we order the two phenomena a regular ascending order by giving each student a rank for a subject so does the other subject for the same student as shown in the next table.

Subjects 1	Weak	Pass	Weak	good	Weak	excellent	V. good
Ordering with repetition	1	4	2	5	3	7	6
Final repetition	2	4	2	5	2	7	6

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We notice that the case “weak” has repeated three times and occupied the places 1,2 and 3, hence, the rank of each $= \frac{1+2+3}{3} = 2$ (it is the arithmetic mean of the numbers 1,2 and 3), and similarly,

Subjects 1	Weak	Pass	good	Pass	Weak	Very good	Pass
Ordering with repetition	1	3	6	4	2	7	5
Final repetition	1.5	4	6	4	1.5	7	4

We notice that the level “weak” has repeated twice and occupied the places 1 and 2, hence, the rank of each $= \frac{1+2}{2} = 1.5$ it is the arithmetic mean of the numbers 1 and 2). Also, the level ”pass” has repeated three times and occupied the places 3,4 and 5, hence, the rank of each $= \frac{3+4+5}{3} = 4$.

We summarize the solution in the table as follows:

x	y	Ranks of x	Ranks of y	D	D²
Weak	Weak	2	1.5	0.5	0.25
Pass	Pass	4	4	0	0
Weak	good	2	6	-4	16
good	Pass	5	4	1	1
Weak	Weak	2	1.5	0.5	0.25
excellent	V. good	7	7	0	0
V. good	Pass	6	4	2	4
Sum					21.5

$$\therefore r = 1 - \frac{6 \sum D^2}{n(n^2 - 1)} = 1 - \frac{6(21.5)}{7(7^2 - 1)} \approx 0.6161 \quad \text{It is a direct correlation.}$$

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Try to solve ➤

In a study about the relation between the students' levels in statistics and mathematics, the degrees of six students have been as follows:

Degrees of statistics(x)	Pass	Very good	excellent	Very good	Pass	Pass
Degrees of mathematics (y)	Good	good	Very good	excellent	good	weak

calculate the Spearman's rank correlation coefficient among the degrees and identify its type.

Example 4 ➤

calculate the Spearman's rank correlation coefficient between x and y through the data of the next table:

x	4	7	8	5	8	12
y	7	6	6	4	6	10

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Solution:

Form the next table:

x	y	Ranks of x	Ranks of y	D	D^2
4	7	6	2	4	16
7	6	4	4	0	0
8	6	2.5	4	-1.5	2.25
5	4	5	6	-1	1
8	6	2.5	4	-1.5	2.25
12	10	1	1	0	0
Sum					21.5

$$\therefore r = 1 - \frac{6 \sum D^2}{n(n^2 - 1)} = 1 - \frac{6(21.5)}{6(6^2 - 1)} \approx 0.3857 \quad \text{The correlation is direct}$$

Critical thinking: Does D^2 vary if we order the two phenomena x and y ascendingly? explain your answer

Try to solve ➤

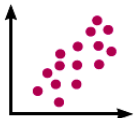
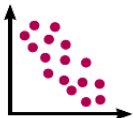
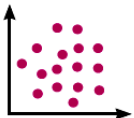
Calculate the Spearman's rank correlation coefficient between x and y and identify its type from the data shown in the next table:

x	10	7	8	7	6	12
y	5	8	7	9	9	10

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EXERCISES

First: Choose the correct answer:

- ① The strongest correlation coefficient of the following is:
a -0.94 b zero c 0.5 d 0.85
- ② The strongest inverse correlation coefficient of the following is:
a -0.2 b -0.5 c -0.7 d -0.8
- ③ the scatter diagram representing inverse correlation is:
a  b  c  d 
- ④ The weakest correlation coefficient of the following is:
a -1.2 b -0.7 c 0.12 d 0.9
- ⑤ One of the next numbers can represent the strongest inverse correlation coefficient between two variables:
a 0.3 b 0.9 c -1.1 d -0.95

Answer the following questions

- ⑥ From the data of the next table

x	12	10	14	11	12	9
y	18	17	23	19	20	15

First: Calculate Spearman's rank correlation coefficient between the two variables x and y.

Second: Calculate Pearson's linear correlation coefficient between x and y.

- ⑦ From the data of the next table

x	7	7	8	3	7	11
y	8	4	12	2	10	11

Calculate Spearman's rank correlation coefficient between the two variables x and y.

- ⑧ From the data of the next table

x	1	3	4	6	7	9
y	6	4	4	3	2	1

Calculate Pearson's correlation coefficient between the values of x and y and show its type.

- ⑨ From the data of the next table

x	6	5	7	8	10	6	7
y	4	7	5	6	8	7	8

Calculate Pearson's correlation coefficient between the values of x and y and show its type.

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- 10 From the data of the next table

x	3	1	6	4	3	8
y	7	4	5	8	6	7

Calculate Spearman's rank correlation coefficient between x and y and show its type.

- 11 From the data of the next table

x	Very good	Very good	good	Weak	Pass	Very good
y	good	Pass	good	excellent	Very good	Pass

Calculate Spearman's rank correlation coefficient between x and y.

- 12 Find Pearson's correlation coefficient between the values of x and y and show its type if:

$$\sum x = 220$$

$$\sum y = 140$$

$$\sum xy = 2658$$

$$\sum x^2 = 5486$$

$$\sum y^2 = 2292$$

$$n = 10$$

- 13 **Trade:** The following table shows a 6 - book group in regard to its price (x) and the sales (y).

Price (x)	Low	Very low	Average	Very high	high	Very high
Sales (y)	high	high	Very high	Low	Average	Low

Calculate Spearman's rank correlation coefficient between the price and sale of the book.

- 14 **Advertisement:** An advertising company has decided to study the relation between its expenditure on advertisement x (in thousand L.E) and its sales y (in thousand units). If the data of the company's eight branches are as follows:

x	19	18	7	10	4	13	15	5
y	12	10	7	9	6	13	14	12

Find the rank correlation coefficient between the expenditure and sales and show its type.

- 15 **Education** The next data represent the marks of ten students in chemistry and biology.

chemistry	60	85	55	90	65	50	80	70	95	75
biology	55	75	50	95	60	65	85	80	90	70

Calculate Pearson's linear correlation coefficient and determine its type.

- 16 **Birth:** In a study to determine the relation between the mother's age and the number of her kids, the following data have been determined:

mother's age	18	20	23	27	29	32	33	35
number of kids	2	1	1	2	3	4	3	5

Calculate spearman's rank correlation relation and determine its type.

Regression

Introduction:

You have previously studied the function and identified its graph. In the previous part, you learned the scatter diagram and you also realized that the objective of graphing it is to identify the nature of the relation between two variables x and y through the data related to both variables. You also learned that the properties of the correlation between two phenomena can be in the form of the following:

Linear Relation

negative Linear Relation

non-Linear Relation

no Relation

In this part, you are going to learn how to identify the regression line equation. The purpose of this study is to help the researcher know the type of the given data and perform correct predictions through them.

Definition of Regression:

Regression is a statistical method by which the value of a variable can be estimated in terms of the other variable.

simple linear regression: in which, the dependent variable (y) depends on one variable (x) through a linear relation.

Regression line equation:

You have previously learned in the analytic geometry the straight line equation whose slope is m and y-intercept of b is $y = m x + b$.

Let its equation be: $y = a + bx$

The most common method to find the best values of a and b is called the least square method.

The least square method:

Of the mentioned previously, we knew that in case of correlation, it is not necessary all the points lie on the regression line. Therefore, there is an error ratio for the points which do not lie on the regression line.

To get the best regression line, the least possible valued deviations should be reduced (the proper regression line should pass or approach to the greatest number of scattering points). if (x, y) is a true point of data and $(x, \hat{y})$ is the point locating off on the regression line ($\hat{y}$ is read as y hat), then the proper regression line when $|y - \hat{y}|$ is the least

for the values of x or when $(y - \hat{y})^2$ is the least and by supposing the regression line equation is $\hat{y} = a_0 x + a_1$

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The required is to determine the values of a_0, a_1 such that the absolute difference is minimum by solving the following two equations

$$\sum y = a_0 \sum x + n a_1 \quad \text{and} \quad \sum xy = a_0 \sum x^2 + a_1 \sum x$$

therefore

$$a_0 = \frac{n \sum x y - \sum x \sum y}{n \sum x^2 - [\sum x]^2}, \quad a_1 = \frac{\sum y \sum x^2 - \sum x \sum x y}{n \sum x^2 - [\sum x]^2}$$

EXAMPLE 1►

The next table illustrates the production of a summer crop (y) of the cultivated land (x) in feddan.

cultivated land (x) in feddan	50	200	110	80	120	74.5	88.9	5.7	11	3.2
production of (y) in kg	140	500	400	300	356	240.5	200.6	33.5	69.8	18.7

First: find the regression line equation.

second: predict the value of the production in kg if the cultivated land is 100 feddans.

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Solution:

First: You can check that

$$\Sigma x = 743,3$$

$$\Sigma y = 2259.1$$

$$\Sigma x^2 = 89017,19$$

$$\Sigma xy = 254489,18$$

then

$$a_1 = \frac{\sum y \sum x^2 - \sum x \sum xy}{n \sum x^2 - [\sum x]^2} = \frac{(2259.1)(89017.19) - (743.3)(254489.18)}{10(89017.19) - (743.3)^2} = 35.35,$$

$$a_0 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - [\sum x]^2} = \frac{10(254489.18) - (743.3)(2259.1)}{10(89017.19) - (743.3)^2} = 2.5637$$

then the regression line has the form $y = 2.5637x + 35.35$

Second: From the regression line equation: $y = 2.5637x + 35.35$

$$\text{at } x = 100 \Rightarrow \hat{y} = 2.5637(100) + 35.35 = 291.72 \text{ kg}$$

EXAMPLE 2 ➤

the following table shows the data about the average price of the oil barrel and the rates of the economic growth in a country within eight years. Required is to:

Price of the oil barrel (x)	36	40	36.2	31.1	29.7	16.3	18.7	14.6
Rates of the economic growth (y)	0.91	3.5	3.2	2.7	2.3	- 1	- 0.9	- 1.6

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First: find the regression line equation of the given data.

second: predict the economic growth when the price of the oil barrel is \$15, then when the price is \$35.

Solution:

x	y	x ²	y ²	x y
36	0.91	1296	0.8281	32.76
40	3.5	1600	12.25	140
36.2	3.2	1310.44	10.24	115.84
31.1	2.7	967.21	7.29	83.97
29.7	2.3	882.09	5.29	68.31
16.3	-1	265.69	1	-16.3
18.7	-0.9	349.69	0.81	-16.83
14.6	-1.6	213.16	2.56	-23.36
222.6	9.11	6884.28	40.2681	384.39

First: From the table data:

$$\begin{aligned}\Sigma y &= 9.11 & \Sigma x &= 222.6 \\ \Sigma x y &= 384.39 & \Sigma x^2 &= 6884.28\end{aligned}$$

then

$$a_0 = \frac{n \sum x y - \sum x \sum y}{n \sum x^2 - [\sum x]^2} = \frac{8(384.39) - (222.6)(9.11)}{8(6884.28) - (222.6)^2} = 0.1896$$
$$a_1 = \frac{\sum y \sum x^2 - \sum x \sum x y}{n \sum x^2 - [\sum x]^2} = \frac{(9.11)(6884.28) - (222.6)(384.39)}{8(6884.28) - (222.6)^2} = -4.1368,$$

then the regression line has the form

$$y = 0.1896x - 4.1368$$

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Second:

When $x = 15$ then : $y = 0.1896(15) - 4.1368 \approx -1.2928$

When $x = 35$ then : $y = 0.1896(35) - 4.1368 \approx 2.4992$

Try to solve ➤

In a study to find the relation between the income (x) and consumption (y) in thousand pounds, the results show that:

$$\begin{array}{lll} n = 40, & \sum x = 120, & \sum y = 100 \\ \sum x^2 = 720, & \sum y^2 = 410, & \sum xy = 516 \end{array}$$

- a) Find the linear correlation coefficient between x and y .
- b) Find the regression line equation.
- c) Predict the value of consumption (y) when the income is 10000 LE.

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EXERCISES

First: choose the correct answer:

- ① the statistical equation of the regression line equation where b is the regression coefficient is:
- ☐ a $\hat{y} = a x + b$ ☐ b $\hat{y} = a + b x$
☐ c $\hat{y} = a y + b$ ☐ d $\hat{y} = a + b y$
- ② if the regression line equation is: $\hat{y} = 2 + 0.5x$ then the value of y expected when $x = 6$ is:
- ☐ a 4 ☐ b 5 ☐ c 7 ☐ d 8
- ③ if the two points (11.5 , 10) and (6.5 , 5) lie on the regression line y on x , then the correlation between x and y is:
- ☐ a direct ☐ b inverse ☐ c perfect ☐ d nihilistic
- ④ if the two points (5 , 13) ,and (14 , 4) lie on the regression line y on x , then all the following points lie on the same line except:
- ☐ a (15 , 5) ☐ b (10 , 8) ☐ c (6 , 12) ☐ d (5 , 13)
- ⑤ if all the points in a scatter diagram lie on a straight line whose slope is negative, then the correlation coefficient between x and y equals:
- ☐ a 1 ☐ b zero ☐ c - 0,5 ☐ d -1
- ⑥ if all the points in a scatter diagram lie on a straight line whose slope is positive, then the correlation coefficient between the two variables equals:
- ☐ a - 1 ☐ b zero ☐ c $\frac{1}{2}$ ☐ d 1

Second: Answer the following questions:

- ⑦ the next table illustrates the relation between the two variables x and y :

x	5	8	10	14	16	20
y	4	6	9	11	12	15

- ☐ a graph the scatter diagram ☐ b find the regression line equation
☐ c predict the value of y when $x = 12$
- ⑧ from the data in the next table:

x	20	33	30	40	13	15	26	25
y	7	8	9	11	4	5	8	9

- ☐ a predict the value of y when $x = 35$
☐ b find the error in y if $x = 30$

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- 9 in a statistical study to find the relation between two variables x and y , we have got the next data:

$n = 10$, $\bar{x} = 8$, $\bar{y} = 10$, $\sum xy = 870$, $\sum x^2 = 665$, $\sum y^2 = 1400$ find:

- a linear correlation coefficient
 - b regression line equation
- 10 if : $\sum x = 30$, $\sum y = 40$, $\sum xy = 162$
 $\sum x^2 = 210$, $\sum y^2 = 304$, $n = 6$ find :
- a regression line equation
 - b linear correlation coefficient between x and y and identify its type

- 11 **sales:** In a market for selling used cars, the sales were as follows:

car age (x)	3	2	1	1	5	6	1	4
Sale price (y)	54	80	74	98	45	40	85	60

Find:

- a Pearson's linear correlation coefficient
 - b Regression line equation
- 12 **Economy:** the next table represents the monthly income (x) and the expenditure (y) for a group of families in hundred pounds:

Income (x)	38	27	39	40	56	66	42	44
expenditure (y)	19	25	20	28	31	38	27	22

- a find Pearson's rank correlation coefficient and identify its type.
 - b find the regression line equation
 - c Estimate the value of the expenditure (y) if the income (x) is 5000 LE
 - d find the error in (y) if (x)= 40
- 13 **Family** to study the relation between the income (y) and consumption (x) in hundred pounds monthly in a city, a 40- family sample has been taken to give the following results:
 $\sum x = 100$, $\sum y = 120$, $\sum xy = 516$, $\sum x^2 = 410$, $\sum y^2 = 720$.
- a find the regression line equation
 - b predict the family income whose monthly consumption is 700 LE.